

Chapter 13

A Need For Safety

Name: _____

1. Define the following

a) Statistics _____

b) Reaction time _____

c) Distraction _____

d) Stimulant _____

e) Impaired _____

f) Depressant _____

g) BAC _____

h) Drink _____

i) Impaired driving _____

j) Trauma _____

k) Whiplash _____

l) Grief _____

m) Therapy _____

n) Maintenance _____

2. In Canada how much more likely will you die in a road collision than on an airplane?
Pg. 234

3. What type of injuries do you get in a car crash? Pg 235

4. Even if alert what are some factors that might affect your ability to stop? Pg 237

5. Why shouldn't you follow too closely? Pg 237

6. List 4 things that slow your reaction time. Pg 237, 240, 241.

7. Are cell phones a problem when driving? Why or Why not? Pg 237

8. What should you do if you feel fatigued when driving? Pg 240

9. Why are stimulants dangerous to take when driving? Pg 240

10. Why does Alcohol slow reaction time? Pg 240

11. What is the legal blood alcohol limit? Pg 241

12. What factors affect your BAC? Pg 241

13. Put the following people in order from drunkest to Sober.

1. A 300lb male drinks 4 drinks in 2 hours.
2. A 90lb female who has 2 drinks in 1 hour.
3. A 200lb 55 year old alcoholic who has 3 drinks in 2 hours
4. A 90lb female who has 2 drinks in 1 hour on a full stomach.

14. Use the following chart for the next question.

BAC Based on Number of Drinks

Body Mass Max. (kg)	Female Number of Drinks						Male Number of Drinks					
	1	2	3	4	5	6	1	2	3	4	5	6
45	0.05	0.10	0.152	0.203	0.253	0.304	0.043	0.087	0.130	0.174	0.217	0.261
57	0.04	0.080	0.120	0.162	0.202	0.244	0.034	0.069	0.103	0.139	0.173	0.209
68	0.034	0.068	0.101	0.135	0.169	0.203	0.029	0.058	0.087	0.116	0.145	0.174
79	0.029	0.058	0.087	0.117	0.146	0.175	0.025	0.050	0.075	0.100	0.125	0.150
91	0.026	0.050	0.076	0.101	0.126	0.152	0.022	0.043	0.065	0.087	0.108	0.130
102	0.022	0.045	0.068	0.091	0.113	0.136	0.019	0.039	0.058	0.078	0.097	0.117
114	0.020	0.041	0.061	0.082	0.101	0.122	0.017	0.035	0.052	0.070	0.087	0.105

a) The body eliminates 0.015BAC units per hour. The legal limit is 0.08. If you drank 4 drinks in 1 hr how long will you have to wait before you can drive. (228)

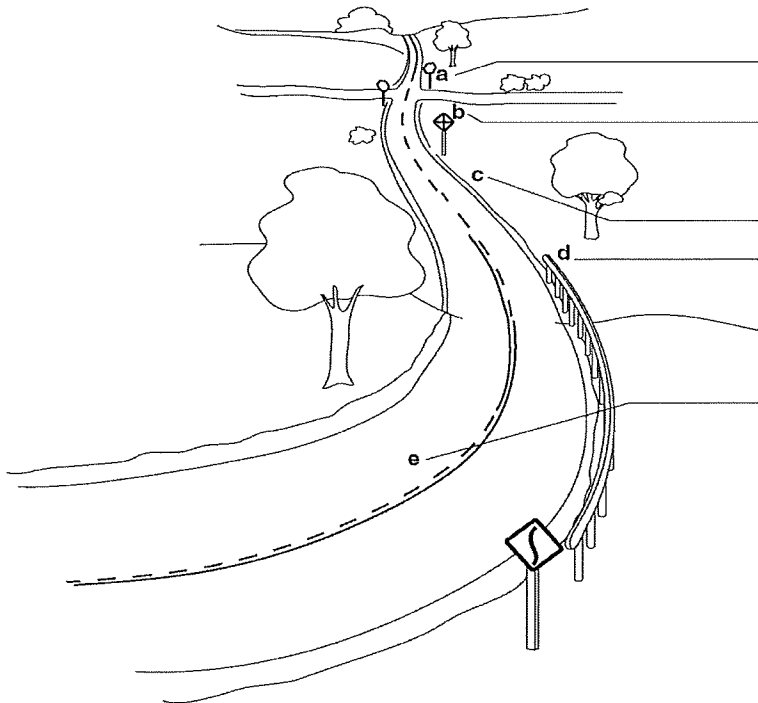
b) How long would it take a 68kg Female to sober up after 6 drinks in an hour?

15. What part of the car will prevent whiplash? (pg 243)

16. List 4 safety features that your car has. (pg 245)

17. Why should you perform regular maintenance on a vehicle? (pg 245)

18. Label the safety features on this road. (pg 247)



Chapter 14: The Nature of Speed

Name: _____

1. Define the following

a) distance _____

b) time _____

c) velocity _____

d) distance/time graphs _____

e) variables _____

f) manipulated variable _____

g) responding variable _____

h) run _____

i) rise _____

j) slope _____

k) braking distance _____

l) stopping distance _____

m) following distance _____

2. What is another word for velocity? Pg 252

3. What is the formula for calculating velocity? Pg 254

4. Rearrange the formula above to solve for distance. Pg 254

5. How does increasing the distance affect the time if the velocity stays the same?(pg 254)

6. How would doubling the velocity while keeping distance the same affect time of travel? Pg 254

7. Cameron drives 300 km in 2 hours. How fast is he traveling? Show your work. Pg 254)

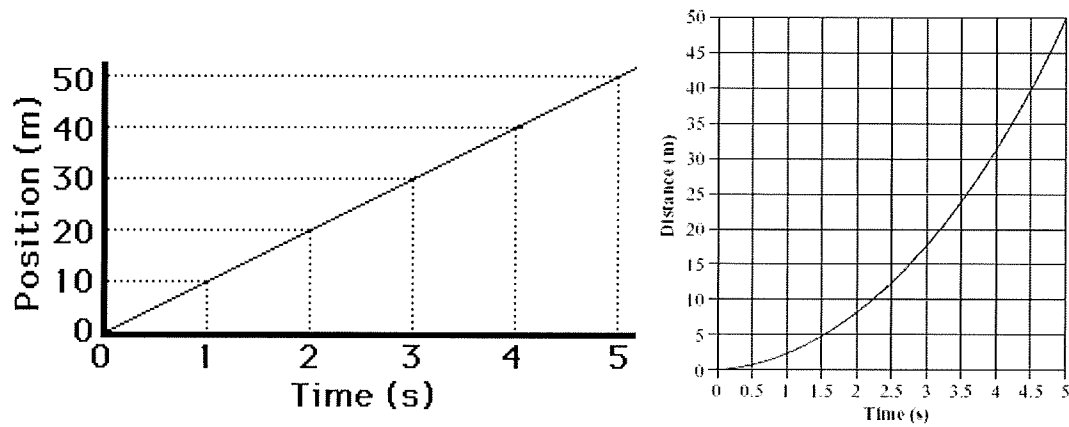
8. Brandon is running to Macs to grab a slurpee at 10 minute break (600s). Round trip to Macs from MCHS is 1600m. Can Brandon make it in time if he runs 10m/s?

9. On a distance vs time graph on what axis does the time go? (Pg 256)

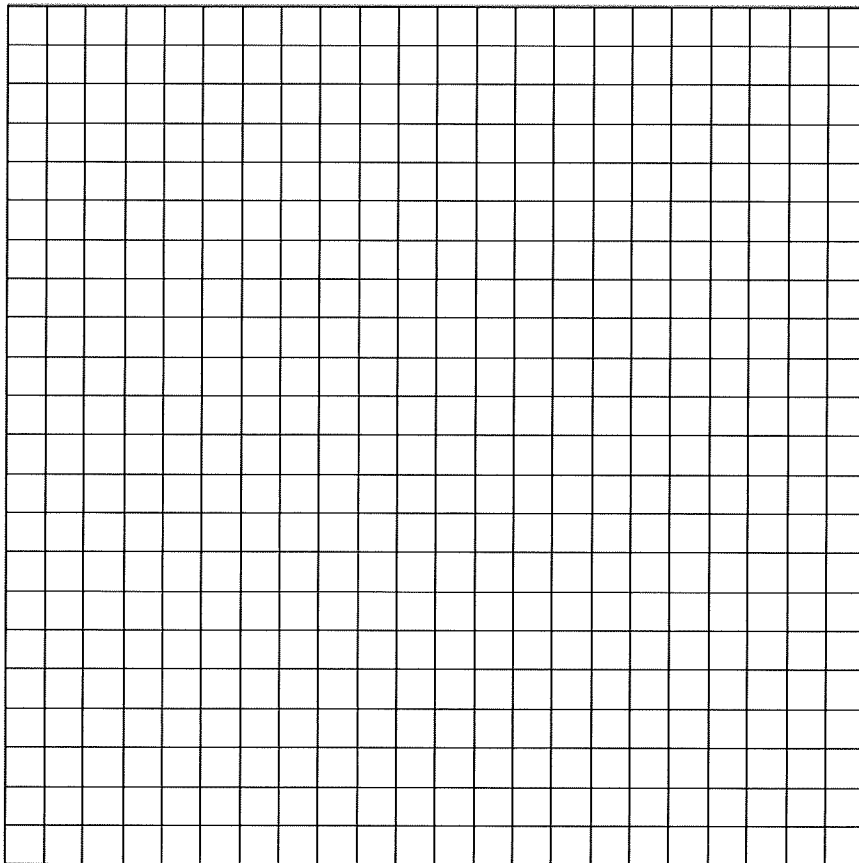
10. On a distance vs time graph what axis does the distance go? (Pg 256)

11. What type of line do we draw through the points on a distance vs time graph? (Pg 256)

12. Circle the following graph that show someone traveling at a constant speed? (pg 257)

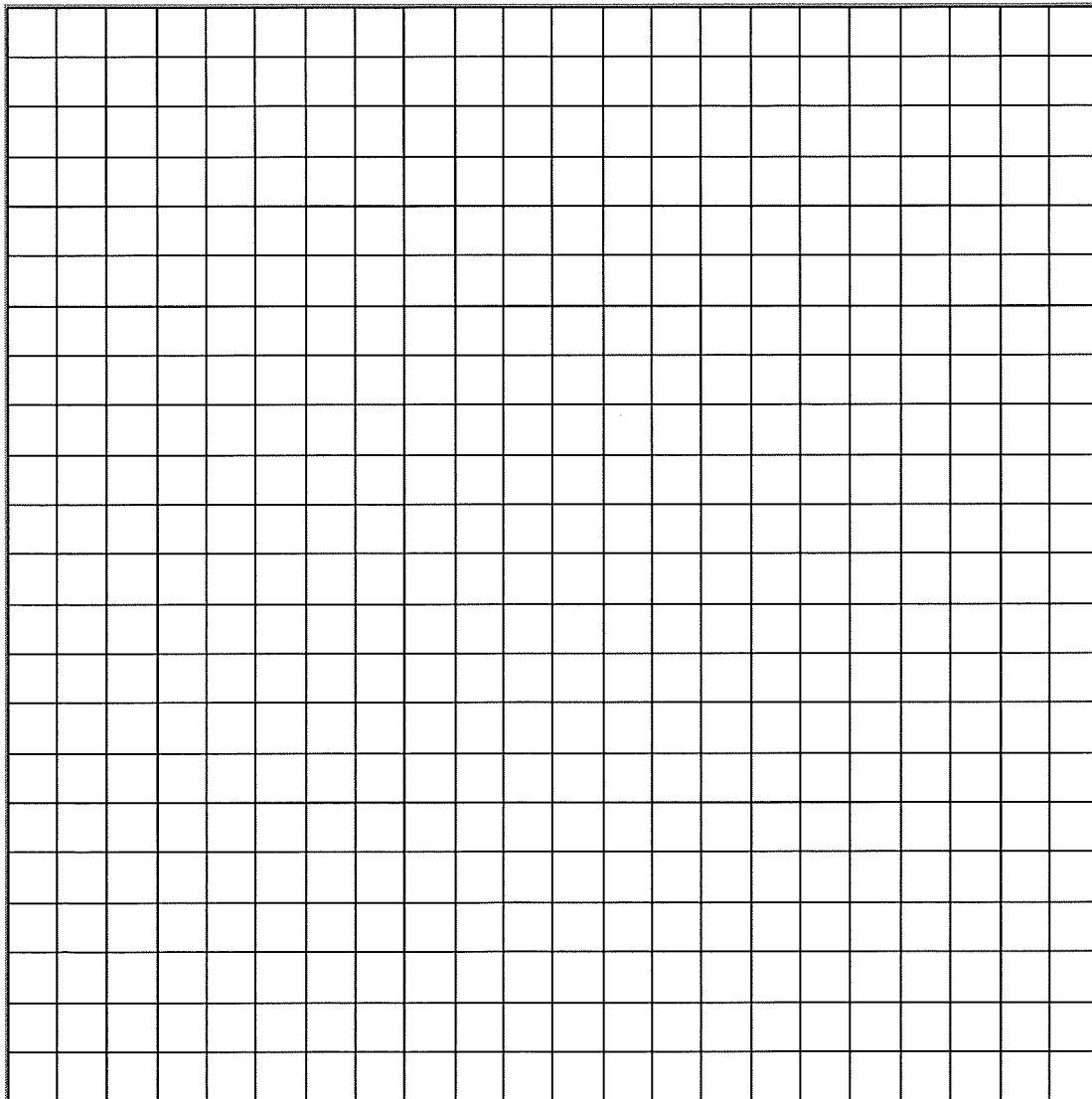


13. Draw a graph rough of the following situation. Tyler runs at a constant speed for 5 seconds. He then stops for 5 seconds and runs again.



14. Graph the following table of values on a graph. Be sure to create a title and to label all your axis.

Time	0s	1s	2s	3s	4s	5s
Distance	0s	2m	4m	6m	8m	10m

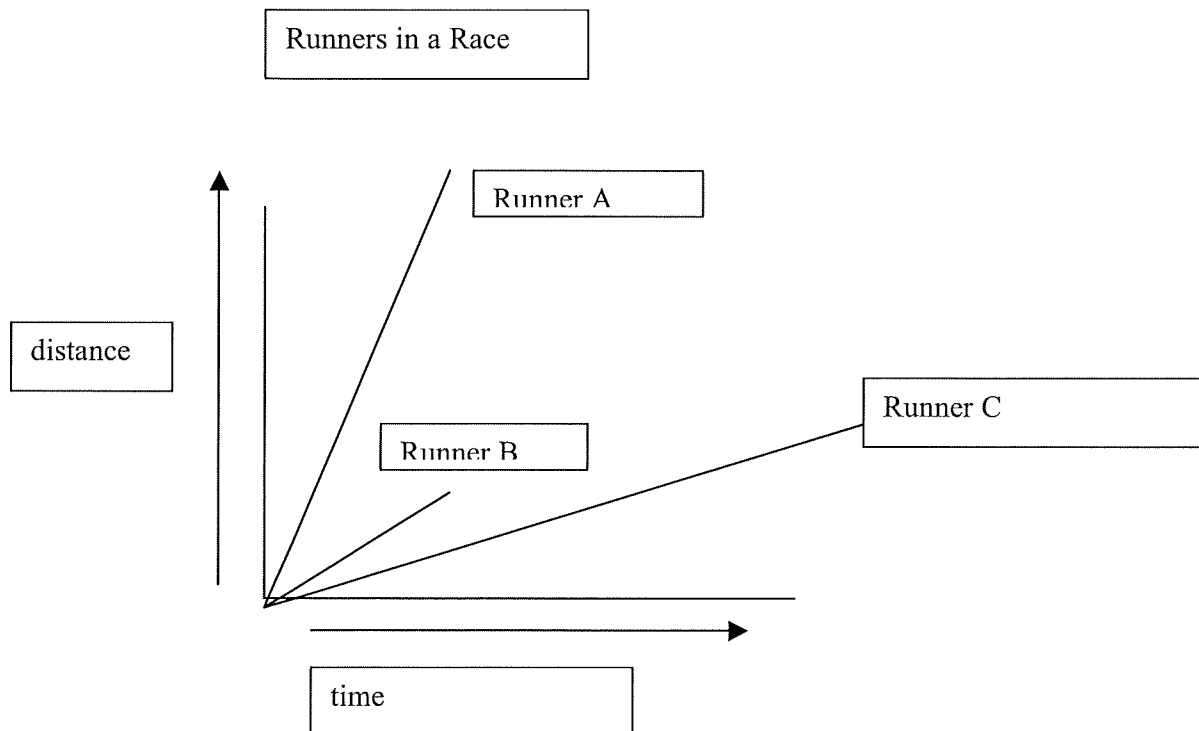


15. Calculate the slope of the object. (Pg 260)

16. What does the slope give us? (Pg 261)

17. What does the horizontal line a distance vs. time graph represent?

18. Use the following graph to answer the next 3 questions.



- Which runner was running the fastest?
- Which runner went the shortest distance?
- Which runner ran for the longest time?

19. Complete the following Chart.

Trip	Distance	Time	Speed
Edmonton to Red Deer		1.35h	110km/h
Jasper to Hinton		0.66h	100km/h
Morinville to Vegas	1966km		120km/h
Morinville to Drumheller	275	2h	
Morinville to Jasper		3.11h	100km/h
Morinville to Edmonton	28km	0.167h (10 min)	

20. What is the difference between Braking distance and Stopping distance? (Pg 264)

21. When should you increase following distance? (Pg 265)

Chapter 15

Name: _____

1. Define the following:

- a. Momentum _____

- b. Force _____

- c. Braking _____

- d. Friction _____

- e. Impulse _____

- f. Law of Conservation of Momentum _____

- g. Total momentum _____

2. When will an object that is not moving begin to move? Pg 270

3. When will an object that is moving stop? (pg 270

4. How do we calculate momentum? Pg 271

5. Fill in the following chart: Pg 271

Object	Mass	Velocity	Momentum
Compact Car	750kg	100km/h	
Adult female on a bike	70kg	30km/h	
Adult female	70kg	5km/h	
Transport truck	2500kg	50km/h	
Transport truck	2500kg	100km/h	

6. Give an example of an object with a very high momentum and one with a very low momentum. Pg 271

7. How is momentum affected in each situation. Pg 271

a. Doubling velocity

b. Doubling mass

c. Doubling both velocity and mass

8. How do brakes work? Pg 272

9. Why don't brakes work as well in wet or icy conditions? Pg 272

10. How do we calculate impulse? Pg 274

11. Solve the following questions

a. A football player collides with another player. The force of the collision is 100N. The collision lasts 2 seconds. What is the impulse?

b. A baseball player feels a frictional force of 400N as she slides into home. What is the impulse if the slide lasts 0.5 seconds?

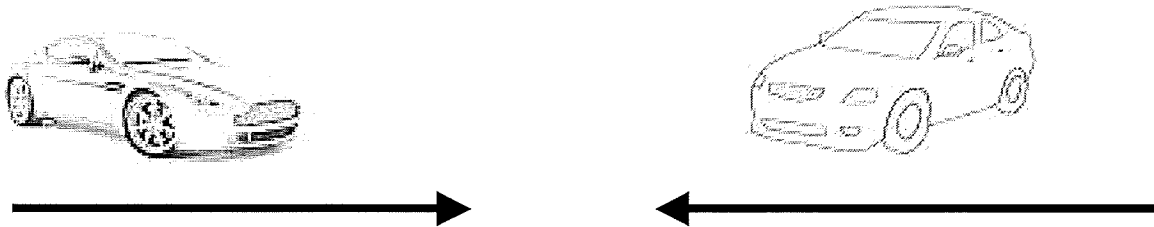
12. How does increasing your stop time protect you from injury? Pg 274

13. Is the force needed to stop a car the same whether you take longer or slower to stop? Explain.

14. In a collision that stops a vehicle, how does the original velocity relate to the force experienced by vehicle and passenger? In other words if a car hits a brick wall how does the speed at which it hits the wall affect the force felt by the passenger? (pg 277)

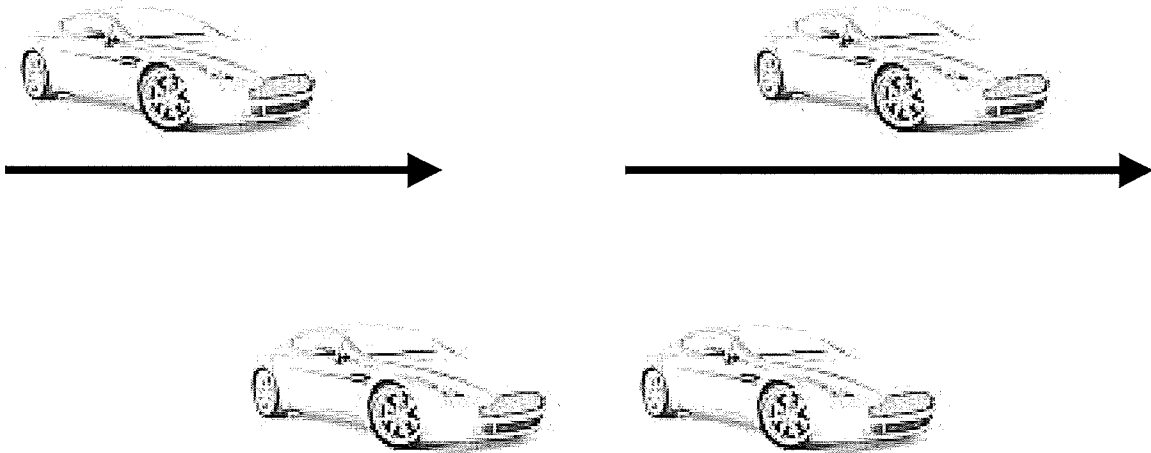
15. Use the following diagrams and determine what happens to momentum Pg 281

A) Two cars of the same size and traveling at the same speed hit in a head on collision.

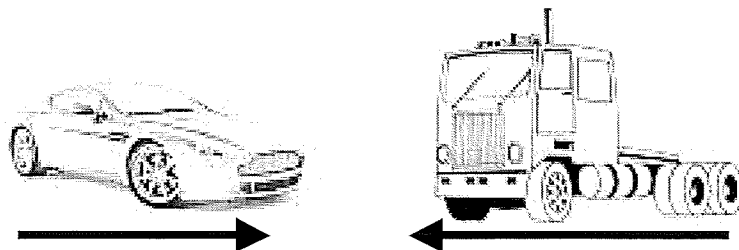


B) Draw the arrows and explain what would happen to momentum after collision.

Two cars traveling in the same direction hit.



C) What would happen in a head on collision between these two vehicles? Assuming both vehicles were traveling at the same speed.



Chapter 16

Name: _____

1. Define the following:

- a. Crash test dummies _____

- b. Crumple zones _____

- c. Restraining features _____

- d. Operational features _____

- e. Structural features _____

- f. Road safety features _____

- g. Crash cushion _____

- h. Guardrails _____

- i. 2-point seat belts _____

- j. lap belts _____

- k. 3-point seat belts _____

- l. shoulder harness _____

- m. infant-only seats _____

- n. child seat _____

- o. convertible child seats _____

- p. booster seats _____

- q. air bag _____

2. What do safety features do on a car? Pg 288

3. Why do vehicles have crumple zones? Pg 288

4. List 4 examples of a restraining feature. Pg 289
5. Why are your rear and side mirrors an example of an operational feature? Pg 289
6. What are the two types of structural features? List an example of each. Pg 289
7. Why are the rigid features of a car surrounding the cabin of the vehicle? Pg 289
8. What is a major cause of injury in a car with no crumple zones? Why? Pg 290
9. List 4 road features that were designed to reduce the force of impact. Pg 291

10. What type of injury would happen if 2-point belts were in the front seat? Pg 292

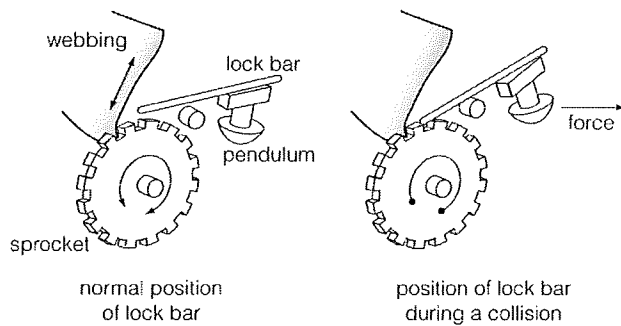
11. Why haven't engineers found a way to restrain the head in a car? Pg 292

12. Label the following picture Using the words webbing and retractor unit. Pg 293



13. Why does the webbing stretch? Pg 293

14. Explain how this structure works. Pg 293



15. Circle the right answer

a. Babies heads are (**larger** / **smaller**) in proportion to their body. Pg 296

16. Why should car seats be facing the rear of the vehicle? Pg 297

17. What type of gas does an air bag fill up with? Pg 298

18. What was the problem with the first generation air bag? Pg 298

19. List 2 features of the third generation air bags. Pg 301

